

EDUCATION

The University of Hong Kong, Computer Science

PhD Student in Robotics, supervised by Jia Pan

Hong Kong

Sept. 2025 - Expected Jun. 2029

Shanghai Jiao Tong University, Robot Institute

Master of Engineer, Mechanical Engineering, supervised by Zhenhua Xiong

Shanghai

Sept. 2022 - Jun. 2025

Shanghai Jiao Tong University, The School of Mechanical Engineering

Bachelor of Engineer (with honors), Mechanical Engineering, supervised by Feifei Chen and Ye Ding

Shanghai

Sept. 2018 - Jun. 2022

EXPERIENCE

LimX Dynamics

Intern working on real-world reinforcement learning for humanoid robot

Shenzhen

Oct. 2025 - Sept. 2026

Xense Robotics

Project Lead, tactile real-world RL

Shanghai

Mar. 2026 - Sept. 2026

University of California, Berkeley, MSC-Lab

Research Intern advised by Prof. Masayoshi Tomizuka, Mingyu Ding

Berkeley

May 2024 - Jan. 2025

AWARDS

- National Scholarship, 2023, 2020, Top 2%
- Shanghai Merit Graduate, 2022, Top 2%
- Shanghai-FANUC Scholarship, 2021, Top 1%
- Zhiyuan Honors Degree, 2022, Top 1.5%
- Agilent Scholarship, 2022, Top 3%
- Zhiyuan Honors Scholarship, 2018, 2019, 2020, 2021, Top 5%

RESEARCH INTERESTS

I am interested in enabling robots to learn, adapt, and interact with the physical world intelligently and safely. My recent interests focus on humanoid robot learning and tactile perception and manipulation, particularly efficient policy adaptation in real-world environments.

- **Humanoid Robot Learning:** Whole-body VLA, Loco-manipulation, Real-World Reinforcement Learning
- **Tactile Manipulation:** VTLA, Failure Reasoning, Diffusion-Guided Adaptation
- **Multi-Robot Systems:** Navigation, Cooperative Manipulation, Motion Planning, Relative Localization
- **Perception and Planning:** Scene Understanding, SLAM, Lightweight Mapping, Autonomous Exploration

PUBLICATIONS (*EQUAL CONTRIBUTION)

- [1] **K. Song***, Y. Wang*, Y. Chen*, T. Ding*, J. Tian, T. Gong, D. Ma, and J. Pan, “PEARS: Physical-Prior-Guided Efficient Adaptation via Failure Reasoning and Diffusion Steering for Tactile Manipulation,” submitted to ICRA 2027
- [2] **K. Song** et al., “RAPID: Efficient Real-World Adaptation of Humanoid Policies via Distillation and Online Reinforcement Learning,” 2026
- [3] **K. Song**, S. Ma, G. Chen, N. Jin, G. Zhao, M. Ding, Z. Xiong, and J. Pan, “CollaBot: Vision-Language Guided Simultaneous Collaborative Manipulation,” *RA-L* under review
- [4] **K. Song**, G. Chen, M. Tomizuka, W. Zhan, Z. Xiong, and M. Ding “P² Explore: Efficient Exploration in Unknown Clustered Environment with Floor Plan Prediction,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025
- [5] **K. Song**, G. Chen, W. Liu, and Z. Xiong, “Multi-Robot Rendezvous in Unknown Environment with Limited Communication,” *IEEE Robotics and Automation Letters (RA-L)*, vol. 9, no. 11, pp. 9478-9485, 2024

- [6] **K. Song**, W. Liu, G. Chen, X. Xu, and Z. Xiong, "FHT-Map: Feature-based Hybrid Topological Map for Relocalization and Path Planning," *IEEE Robotics and Automation Letters (RA-L)*, vol. 9, no. 6, pp. 5401-5408, 2024
- [7] Z. Li, W. Zhang, Y. Wang, **K. Song**, X. Zhou, and D. Ma, "GLOTouch: Global-to-Local Haptic Perception Using a Parallel Gripper for Object Search, Recognition, and Grasping Without External Vision," submitted to ICRA 2027
- [8] G. Chen, **K. Song**, W. Liu, W. Ma, and Z. Xiong, "A Novel Dynamic Localization Graph for Efficient Relative Localization," *IEEE Transactions on Instrumentation and Measurement*, Vol. 75, 2026
- [9] W. Liu, M. Ren, **K. Song**, M. Wang, and Z. Xiong, "A Novel Planning Framework for Complex Flipping Manipulation of Multiple Mobile Manipulators," *IEEE Robotics and Automation Letters (RA-L)*, Vol. 10, pp. 5162-5169, 2025
- [10] M. Ren, W. Liu, **K. Song**, L. Shi, and Z. Xiong, "Containment Control of Multi-Robot Systems with Non-uniform Time-varying Delays," *IEEE Transactions on Robotics (T-RO)*, vol. 41, pp. 1657-1672, 2025
- [11] G. Chen, **K. Song**, X. Xu, W. Liu, and Z. Xiong, "RHAML: Rendezvous-based Hierarchical Architecture for Mutual Localization," *IEEE Robotics and Automation Letters (RA-L)*, vol. 9, no. 7, pp. 6440-6447, 2024
- [12] W. Liu, M. Ren, **K. Song**, M. Wang, and Z. Xiong, "Distributed Motion Control of Multiple Mobile Manipulator System with Disturbance and Communication Delay," *Arxiv*, 2024

RESEARCH EXPERIENCE

RAPID: Real-World Humanoid Policy Adaptation [2]

Sept. 2025 - Sept. 2026, Shenzhen

Reinforcement Learning Intern, advised by **Jia Pan, Wei Zhang, and Hua Chen**

LimX Dynamics

- Developed **RAPID**, a three-stage framework for efficient real-world adaptation of humanoid policies: distilling a frozen GR00T-N1.7 vision-language-action teacher into a compact task policy, improving it through on-policy distillation with masked delta action alignment, and fine-tuning it with Humanoid Q-Learning using human corrective interventions.
- Validated RAPID on the LimX Oli humanoid in Pick Fruit and Hang Cup tasks, improving success rates from 37.04% to 88.89% and from 15.00% to 80.00%, respectively, with about one hour of post-training per task and lightweight deployment independent of the VLA teacher.

PEARS: Physical-Prior-Guided Tactile Policy Adaptation [1]

Mar. 2026 - Sept. 2026, Shanghai

Project Lead, advised by **Jia Pan and Daolin Ma**

Xense Robotics

- Developed **PEARS**, a sample-efficient online adaptation framework for contact-rich tactile manipulation that combines physics-guided force reasoning, VLM-based failure diagnosis from visual outcomes and tactile histories, and tactile-conditioned diffusion steering RL over the latent noise of a frozen GR00T-N1.7 policy.
- Implemented stage-aware hybrid force-position control to update and enforce task-specific contact-force bounds across grasping, wiping, twisting, and liquid-transfer contact stages. Improved simulation success rates by 12.4–37.4 percentage points over the strongest per-task baselines and achieved 19/20 Whiteboard Erasing and 18/20 Pipette Liquid Aspiration successes in real-world tests.

CollaBot: Vision-Language Guided Collaborative Manipulation [3]

Sept. 2024 - Oct. 2025, Hong Kong

Research Assistant, advised by **Jia Pan**

The University of Hong Kong

- Developed **CollaBot**, a framework for simultaneous collaborative manipulation of large objects with multiple mobile manipulators. Integrated SEEM-based target-object point-cloud extraction, LLM/VLM-guided motion-constraint inference, and local-to-global collaborative grasp planning.
- Built **LoGNet** for local large-object grasp pose generation and used VLM reasoning to allocate globally coordinated grasp regions across robots. Designed a two-stage motion-planning pipeline to generate collision-free trajectories under closed-chain end-effector constraints, achieving 77.0% overall success across 300 simulated trials and validating real-world deployability.

Exploration in Clustered Environment with Floor Plan Prediction [4]

May 2024 - Sept. 2024, Berkeley, CA

Mechanical Systems Control Lab (MSC-Lab), Supervisor: **Masayoshi Tomizuka**

University of California, Berkeley

- Implemented **floor plan prediction** in the task of indoor environment exploration. 2D maps of scenes in KTH floor plan dataset are collected for the training process. FPUNet (Floor Plan UNet) is used for floor plan prediction. Then, the local predicted maps are merged for **global predicted map**. Room segmentation and their topology are extracted to provide a high-level guidance for exploration.

Lightweight Mapping for Multi-Robot Localization [6][11][5]

Jan. 2023 - Aug. 2024, Shanghai, China

Robot Control and Machine Vision Lab, Supervisor: **Zhenhua Xiong**

Shanghai Jiao Tong University

- Developed a series of work from **lightweight mapping** to map-based multi-robot mutual localization. Proposed FHT-Map, a feature-based hybrid topological map that introduces lightweight support nodes to reduce storage requirements while preserving relocalization and path-planning capability.

- Extended the lightweight topological representation to multi-robot mutual localization and rendezvous: robots share compact topological structures for relative-pose estimation under limited communication, refine robot poses with RHAML, and select rendezvous points from the merged topological map.

Motion Planning for Multiple Mobile Manipulator System [9][12] Jun. 2023 - Jun. 2024, Shanghai, China

Robot Control and Machine Vision Lab, Supervisor: **Zhenhua Xiong**

Shanghai Jiao Tong University

- Proposed a novel planning framework for **complex flipping manipulation** by incorporating platform motions and regrasping. Two types of trajectories, mobile manipulator planning and regrasping planning, are classified and assigned different priorities for various tasks. Comprehensive experiments emphasize the significance of proposed planner in extending the capabilities of multiple mobile manipulator systems in complex tasks

TEACHING EXPERIENCE

- **Teaching Assistant:** COMP7308 *Introduction to unmanned systems* Fall, 2025, The University of Hong Kong
- **Teaching Assistant:** ME3403 *Introduction to Robotics* Spring, 2024, Shanghai Jiao Tong University
- **Teaching Assistant:** ME4409 *Multi-robot System and Control* Fall, 2024, Shanghai Jiao Tong University

SKILLS

- **Language:** Mandarin Chinese (Native), English (Fluent, TOEFL 104)
- **Knowledge:** VLA, VLM, SLAM, Computer Vision, Deep Learning, Reinforcement Learning
- **Programming:** Proficient in Python, Matlab, Linux, and ROS; familiar with C++, Git
- **Libraries:** Proficient in Numpy, PyTorch, Ceres, OpenCV, PCL, MoveBase; familiar with Moveit, Pandas

ACADEMIC SERVICES

- **Journal Reviewer:** IEEE Transactions on Robotics (T-RO), IEEE Robotics and Automation Letters (RA-L), Robotics and Autonomous Systems (RAS)
- **Conference Reviewer:** IEEE International Conference on Robotics and Automation (ICRA), The Association for the Advancement of Artificial Intelligence (AAAI), IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Advanced Intelligent Mechatronics (AIM)